TEXORA · SESSIONAL NOTES · PRACTICAL STUDY SHEET

Practical Study on the Operation of Weaving Machinery

The loom at work: how a sized warp is gaited and drawn in, how the machine is checked and started, what each motion does through one full revolution — and how to observe, time and record a running loom so that the report shows the machine as it actually ran, not as the book says it should.

TE-2104 · WEAVING TECHNOLOGY SESSIONAL · STUDY 2 · LEVEL 2 TERM 1

1. Object and method of the study

Object: to study the operation of the weaving machine (the lab loom — note its type: shuttle or shuttleless, and its shedding system) through a complete working routine: identification of parts, gaiting and drawing-in, pre-start checks, starting, observation of the motions in sequence, measurement of speed and cloth particulars, and recognition of the common stoppages and faults.

Safety first — the rules of this study: never reach across a running loom; never put a hand near the shuttle, the reed or the take-up rollers while the machine can move; when the loom is to be turned by hand, the power is off and one person only turns the hand wheel while others keep clear. A shuttle leaving the loom is the classic weaving-shed accident; stand to the side of the sley, never in front of a shuttle box, during observation.

2. The machine — parts to identify

- **Warp line** — weaver's beam, back rest, lease rods, drop wires (warp stop motion), heald (harness) shafts with healds, reed in the sley, cloth fell, breast beam, sand roller/pressure roller and cloth roll.
- **Shedding system** — tappet (cam) shaft and tappets, or the dobby head above the loom; find how the shafts are lifted and lowered, and the pattern arrangement fitted.
- **Picking system** — on a shuttle loom: picking stick, picker, shuttle box at each end of the sley, and the shuttle with its pirn. On a shuttleless machine, identify its weft insertion system (rapier, projectile, air-jet or water-jet) instead.
- **Sley and beat-up** — the sley carrying the reed, driven by the crank shaft; the crank's revolution is the loom cycle: one pick per revolution.

- **Take-up and let-off** — the take-up motion (find its change wheel/ratchet, which sets picks per inch) and the let-off motion governing the weaver's beam (friction or positive/negative let-off).
- **Stop motions and protectors** — warp stop motion (drop wires), weft stop motion (centre weft fork), loose-reed/fast-reed warp protector or shuttle protector, and the brake.
- **Temples** — the ring or roller temples at both selvedges holding the cloth out to near reed width.

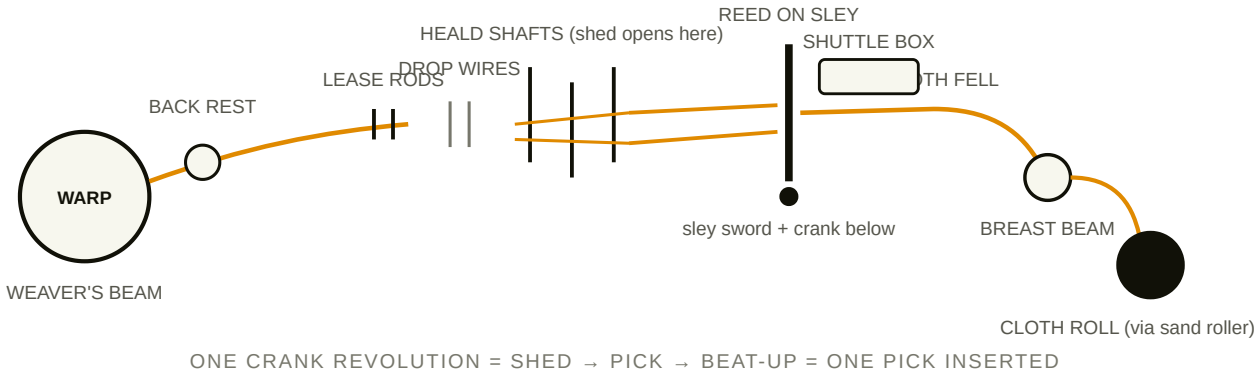


Figure 1 — Side view of the loom: the warp line from weaver's beam to cloth roll. Every end passes, in order, over the back rest, between the lease rods, through a drop wire and a heald eye, and through a dent of the reed — the drawing-in sequence of Section 3.

3. Gaiting and drawing-in — what to observe

1. **Mounting the beam** — the weaver's beam is set in its bearings and the warp sheet brought forward over the back rest; observe how the sheet is kept flat and untwisted.
2. **Lease** — the lease rods preserve the cross (the 1-and-1 order of ends) made at sizing; note what happens to finding a broken end if the cross is lost.
3. **Drawing-in** — each end is drawn, in order, through its drop wire, through the eye of its heald on the shaft assigned by the draft, and through its dent of the reed (the denting plan sets ends per dent). Record the draft in use (e.g. straight draft over the shafts) and the ends per dent.
4. **Tying-in / gaiting** — where a new warp replaces an identical one, ends may be knotted through (tying-in machine or by hand) instead of drawn afresh; note which method the lab uses.
5. **Starting the cloth** — a header is woven, the fell is formed, temples are set, and the take-up is checked by counting the first picks per inch against the setting.

4. Pre-start checks and starting

- Warp sheet free of crossed or slack ends; shed opens cleanly when the loom is turned by hand through two full revolutions.
- Shuttle (or insertion element) boxes correctly; on a shuttle loom the shuttle is sound, the pirn full and correctly wound, picker and check strap in order.
- Reed secure in the sley; temples set at the selvedges; cloth path to the cloth roll unobstructed.
- Stop motions tested: simulate a broken end (lift a drop wire's support / remove an end's tension) and a missing pick, and confirm the loom knocks off.
- Only then: power start, and observe the first minutes for end breaks — a loom that breaks ends repeatedly at start is telling you the warp, the shed setting or the temples need attention, in that order of suspicion.

5. Observing one full cycle

Turn the loom slowly by hand (power off) and record the crank-shaft positions at which events occur. The crank position at which the sley reaches its front-most point and beats up the pick is taken as 0° (front centre). Then check, in order:

EVENT	WHAT TO OBSERVE AND RECORD
Beat-up	Sley at front centre (0°): the reed pushes the newly inserted pick into the fell
Shedding	The shafts change and the shed is fully open in good time before the pick is launched; note the shed timing set on the machine (early/normal/late) and the shed size (clear shed for the shuttle/insertion element)
Picking	The instant of pick launch and the flight of the shuttle through the open shed; it must arrive and be checked in the opposite box before the shed closes on it
Take-up	The small forward movement of the cloth at each pick (continuous or intermittent), governed by the take-up change wheel that sets picks per inch
Let-off	The weaver's beam yields warp as the fell advances; note whether warp tension stays even as the beam diameter falls

6. Measurements to take on the running loom

MEASUREMENT	METHOD AND RECORD
Loom speed	Count crank revolutions (picks) in one minute with a counter or stopwatch = picks/min

Picks per inch in cloth	Count picks in a measured 3-inch length of cloth with a pick glass; divide by 3; compare with the take-up setting
Reed particulars	Reed count (dents per unit length) and reed width from the reed or the loom plate; ends per dent from the denting plan
Warp particulars	Total ends, warp count, and the width in reed
Cloth particulars	Width of cloth on the loom and off-loom (after relaxation) — the difference is the contraction
Efficiency	From the worked check: picks actually inserted in a timed period ÷ picks the speed promises

WORKED CHECKS — EFFICIENCY AND PICKS PER INCH

Efficiency: loom speed 180 picks/min observed over 30 min promises 180 × 30 = 5,400 picks. The pick counter advanced 4,590 picks in that period. Efficiency = 4,590 ÷ 5,400 × 100 = **85%** — the missing 15% is stoppage time (warp and weft breaks, gaiting adjustments).

Picks per inch: 174 picks counted in 3 inches of cloth → 174 ÷ 3 = **58 picks/inch**. If the take-up was set for 60, the cloth is 2 picks/inch light — record the discrepancy; do not silently "correct" an observation to match a setting.

7. Stoppages and faults to recognise

- **Warp break** — drop wire falls, warp stop motion knocks the loom off. Observe how the broken end is found (at the lease) and mended.
- **Weft break / missing pick** — the weft fork detects no pick at the receiving side and stops the loom; a pick woven with a broken tail gives a **broken pick** defect in the cloth.
- **Shuttle trap / smash** — if the shuttle fails to clear the shed, the protector (loose reed or shuttle protector) throws the loom off before the sley beats the trapped shuttle through the warp; a smash means many ends broken at once.
- **Cloth faults on the run** — look for reed marks (fine lines along the warp), temple marks at the selvedges, thick/thin weft bars across the cloth, and oil stains; each has a machine-side cause you should name.

8. What the lab report must contain

SECTION	CONTENT
Machine identity	Maker/type, insertion system, shedding system, nominal width, speed range

Parts record	The parts of Section 2 identified on the machine, with the warp-line sketch (Figure 1 as model)
Gaiting record	Draft, denting plan, lease and tying/drawing-in method used
Cycle record	The hand-turned cycle observations of Section 5, with crank positions as found
Measurements	Speed, picks per inch, reed and cloth particulars, efficiency — all measured, none copied
Stoppages	Every stoppage during the observation period, its cause and the remedy applied

9. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. Why is the loom always turned by hand through a full cycle after gaiting, before it is started on power?
2. Distinguish the warp stop motion from the weft stop motion: what each detects, and how each stops the loom.
3. Over 30 minutes a loom running at 180 picks/min inserted 4,590 picks. Find its efficiency and account for the balance.
4. What is drawing-in, and what two plans govern it?
5. At what crank position does beat-up occur, and why are all other loom timings referred to that point?
6. What do the temples do, and what two faults follow if they are removed or badly set?

Answer key

1. To prove the machine mechanically before it can do damage: the hand-turned cycle shows the shed opening fully and in time, the shuttle passing freely, no crossed or trapped ends, and the reed clearing everything. A fault found at hand speed costs a minute; the same fault at full speed is a smash with dozens of broken ends.
2. The **warp stop motion** watches the warp: every end carries a drop wire; when an end breaks its wire falls and completes the knock-off, stopping the loom with the broken end still findable. The **weft stop motion** watches the pick: the centre weft fork feels for the weft at the receiving side each pick; a missing or broken pick lets the fork pass and the loom stops within a pick or two, before a length of cloth is woven without weft.

3. Theoretical picks = 180 × 30 = 5,400. Efficiency = 4,590 ÷ 5,400 = **85%**. The balance (15%) is the time the loom stood stopped — warp and weft break repairs and adjustments during the period.
4. Drawing-in is passing every warp end, in order, through its drop wire, its heald eye and its reed dent before weaving. It is governed by the **draft** (which shaft each end is drawn on, fixing the weave interlacing) and the **denting plan** (how many ends go through each dent, fixing ends per unit width).
5. Beat-up occurs at **front centre, 0°** of the crank revolution — the sley's foremost position. All timings (shed change, pick launch) are set as crank angles from this point because beat-up is the fixed, easily found event of the cycle to which every other motion must be phased.
6. The temples grip the cloth at both selvedges and hold it out to nearly the reed width against the contraction the weaving tension causes. Without them the cloth draws in, the outer ends chafe and break against the reed, and the reed itself can be damaged by the narrowing fell — while the cloth produced is narrower and uneven at the edges.

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